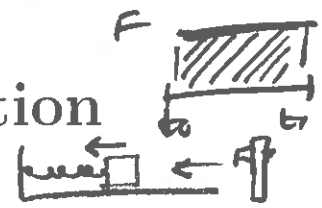
Lecture 24

The Dirac Delta Function

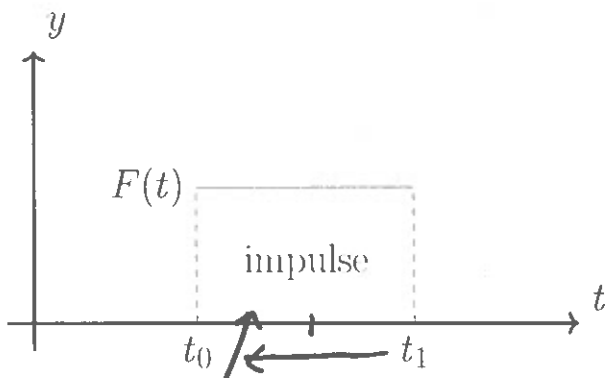


Suppose we strike a mass-spring oscillator with a hammer. Some amount of force is imparted into the system over a time interval $[t_0, t_1]$. If $F(t)$ is the force imparted at time t, then the **impulse** is

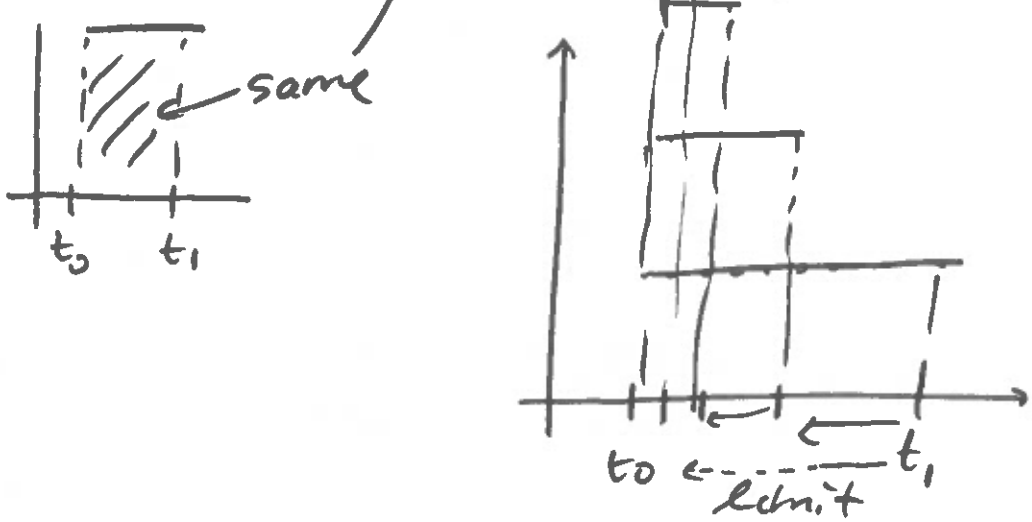
momentum

$$\int_{t_0}^{t_1} F(t)\, dt = \int_{t_0}^{t_1} m\frac{dv}{dt}\, dt = m \cdot v\Big|_{t_0}^{t_1} = mv(t_1) - mv(t_0)$$

number

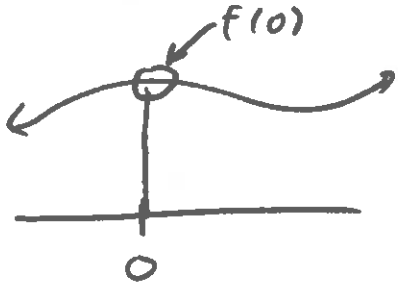


We can note that this impulse is some fixed number and by shortening the time interval and scaling the force to keep the resulting integral constant we can obtain a sequence of functions with a similar graph to that above:



The "limit" of this sequence of functions we define as follows:

Definition: The "function" (distribution) $\delta(t)$ given by



$$\delta(t) = \begin{cases} 0, & t \neq 0 \\ \infty, & t = 0 \end{cases}$$

where

$$\boxed{\int_{-\infty}^{\infty} f(t)\delta(t)\, dt = f(0)}$$

extracts a single value of f.

is called the **Dirac-Delta function**.

Note: $\displaystyle\int_{-\infty}^{\infty} f(t)\delta(t-a)\, dt = f(a)$ for $f(t)$ continuous in an interval around $t = a$ and $\boxed{\displaystyle\int_{-\infty}^{\infty} 1 \cdot \delta(t)\, dt = 1.}$

We have a new function and so let's find its Laplace transform!

Note that for $a \geq 0$,

$$\begin{cases} 0, & t \neq a \\ \infty, & t = a \end{cases}$$

$$\mathscr{L}\{\delta(t-a)\} = \int_0^{\infty} e^{-st}\delta(t-a)\, dt = \int_{-\infty}^{\infty} e^{-st}\delta(t-a)\, dt = e^{-st}\big|_{t=a} = e^{-as}$$

and so $\boxed{\mathscr{L}\{\delta(t-a)\} = e^{-as}}$ and $\boxed{\mathscr{L}\{\delta(t)\} = 1.}$ $\int_{-\infty}^{0} e^{-st}\delta(t-a)\,dt = 0$

As $\mathscr{L}\{\delta(t)\} = 1$ we have that for any $f(t)$ $f*0=0$, $a+0=a$, $a\cdot 1 = a$ } identities

$$\mathscr{L}\{f(t)\} = \mathscr{L}\{f(t)\} \cdot 1 = \mathscr{L}\{f(t)\}\,\mathscr{L}\{\delta(t)\} = \mathscr{L}\{f * \delta\}$$

and thus $f * \delta = f$.

$$\delta * f = \int_0^t \underbrace{\delta(t-v)}_{=0,\ t\neq v} f(v)\, dv = \int_{-\infty}^{\infty} \delta(t-v) f(v)\, dv = f(v)\big|_{v=t} = f(t)$$

Finally, we note that because $\delta(t-a) = 0$ for $t \neq a$ we have

accumulation

$$g(t) = \int_{-\infty}^{t} \underbrace{\delta(x-a)}_{=0\ x\neq a}\, dx = \begin{cases} 0, & t < a \\ 1, & t > a \end{cases} = \underline{\underline{u(t-a)}} \overset{\text{defn}}{=} \begin{cases} 0, & t<a \\ 1, & t>a \end{cases}$$

a is not in $(-\infty, t)$

a in $(-\infty, t)$

and so

$$\frac{d}{dt}\int_{-\infty}^{t} \delta(x-a)\, dx = \boxed{\underbrace{\delta(t)}_{t-a} = u'(t-a)}$$

$$\boxed{u'(t-a) = \delta(t-a)}$$

Example: Evaluate the following

- $\displaystyle\int_{-\infty}^{\infty} \cos(2t)\underbrace{\delta(t)}_{=0\ t\neq 0}\, dt = \cos(2t)\big|_{t=0} = \cos(0) = 1$

$$\boxed{\mathcal{L}\{\delta(t-a)\} = e^{-as}}$$

- $\mathcal{L}\{t^3\delta(t-3)\} = (-1)^3 \dfrac{d^3}{ds^3}\mathcal{L}\{\delta(t-3)\} = (-1)^3 \dfrac{d^3}{ds^3}(e^{-3s})$

$$\boxed{\mathcal{L}\{t^n f(t)\} = (-1)^n \frac{d^n}{ds^n} F(s)}$$

$$= (-1)(-3)^3 e^{-3s} = 27e^{-3s}$$

- $\mathcal{L}\{e^t\delta(t-3)\} = \mathcal{L}\{\delta(t-3)\}\big|_{s\to s-1} = e^{-3s}\big|_{s\to s-1}$

$$\mathcal{L}\{e^{at}f(t)\} = \mathcal{L}\{f(t)\}\big|_{s\to s-a}$$

$$= \boxed{e^{-3(s-1)} = e^{-3s}e^{3}}$$

What is $\mathcal{L}\{\delta(t-a)f(t)\} = \int_0^{\infty} e^{-st}\delta(t-a)f(t)\,dt = e^{-st}f(t)\big|_{t=a}$ $\quad a \geq 0$

$$= \underbrace{e^{-as}f(a)}_{\text{losing information}}$$

Example: Solve the IVP

impulse at $t=\pi$

$$y'' + 2y' + 2y = \delta(t-\pi), \quad y(0) = 1, \quad y'(0) = 1$$

Apply a transform

$$[s^2Y(s) - s\cancel{y(0)}^{1} - \cancel{y'(0)}^{1}] + 2[sY(s) - \cancel{y(0)}^{1}] + 2Y(s) = e^{-\pi s}$$

$$\Rightarrow Y(s)(s^2+2s+2) - s - 1 - 2 = e^{-\pi s}$$

$$\Rightarrow Y(s)(s^2+2s+2) = e^{-\pi s} + (s+3)$$

$s+1+2$

$$\Rightarrow \quad Y(s) = \frac{e^{-\pi s}}{s^2+2s+2} + \frac{s+3}{s^2+2s+2}$$

$(s^2+2s+1)-1+2$

$$= \frac{e^{-\pi s}}{(s+1)^2+1} + \frac{(s+1)+2}{(s+1)^2+1}$$

$$= \frac{e^{-\pi s}}{(s+1)^2+1} + \frac{s+1}{(s+1)^2+1} + \frac{2}{(s+1)^2+1}$$

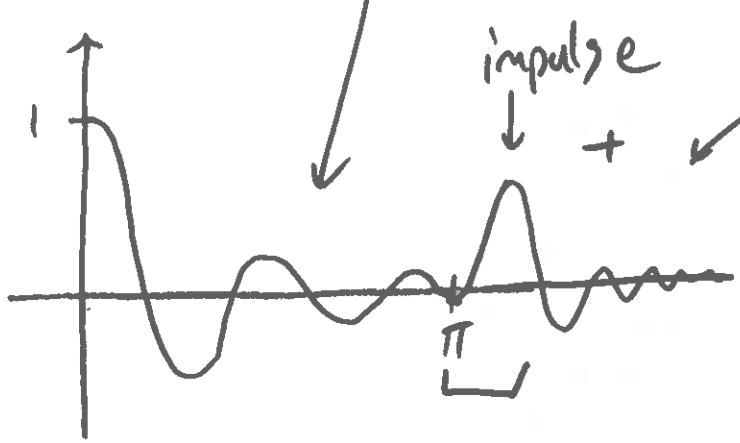
$e^{-\pi s} \cdot \frac{1}{(s+1)^2+1}$

$a=-1 \quad b=1$

$$y(t) = [e^{-t}\cos(t) + 2e^{-t}\sin(t)] + u(t-\pi)(e^{-t}\sin(t)|_{t\to t-\pi})$$

$$\mathcal{L}^{-1}\{e^{-as}F(s)\} = u(t-a)f(t-a)$$

$$\Rightarrow y(t) = [e^{-t}\cos(t) + 2e^{-t}\sin(t)] + u(t-\pi)(e^{-(t-\pi)}\sin(t-\pi))$$



If we consider the IVP

$Y(s)(as^2+bs+c) = 1$

$$ay'' + by' + cy = \delta(t), \quad y(0) = 0, \quad y'(0) = 0$$

then as $\mathscr{L}\{\delta(t)\} = 1$ we have that

transfer function

$$Y(s) = \frac{1}{as^2 + bs + c} = H(s)$$

and thus $\boxed{y(t) = h(t)}$ is the solution. That is, the impulse response function is the particular solution to the constant coefficient equation with forcing function equal to the delta function.

Example: Solve the IVP

$$y'' + y = \delta(t), \quad \boxed{y(0) = 0, \quad y'(0) = 0}$$

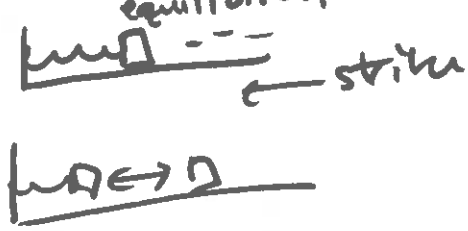
$\Rightarrow [s^2Y(s) - s\cancel{y(0)}^{0} - \cancel{y'(0)}^{0}] + Y(s) = 1$

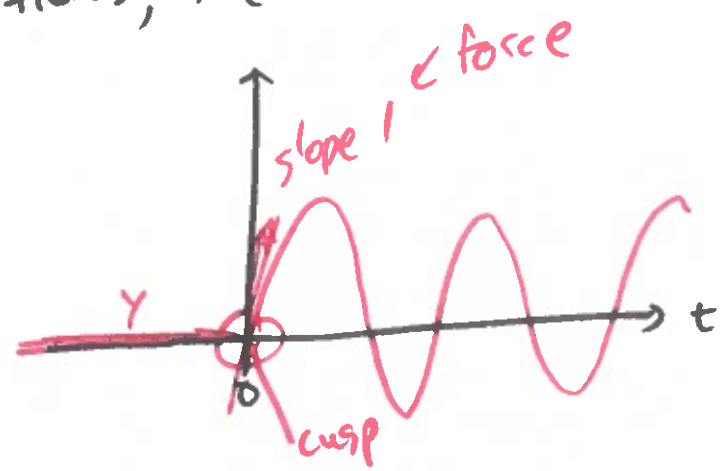
$\Rightarrow Y(s)(s^2+1) = 1 \Rightarrow Y(s) = \dfrac{1}{s^2+1}$

$\Rightarrow y(t) = \sin(t)$ — $y(0) = \sin(0) = 0$, $y'(0) = \cos(0) = 1 \neq 0$

This does not satisfy the conditions!!

If the impulse occurs at the same value as the initial conditions, the solution may not satisfy the conditions.





Example: Solve the IVP

$$\mathcal{L}\{e^t\delta(t-1)\} = \mathcal{L}\{\delta(t-1)\}\big|_{s\to s-1} = e^{-s}\big|_{s\to s-1}$$

$$y'' + 6y' + 5y = e^t\delta(t-1), \quad y(0) = 0, \quad y'(0) = 2$$

$$[s^2Y(s) - sy(0) - y'(0)] + 6[sY(s) - y(0)] + 5Y(s) = e^{-(s-1)} = e^{-s}\cdot e$$

$$Y(s)(s^2+6s+5) - 2 = e^{-s}\cdot e$$

$$\Rightarrow Y(s)(s^2+6s+5) = e^{-s}\cdot e + 2$$

$$\Rightarrow Y(s) = \frac{2}{s^2+6s+5} + e^{-s}\cdot e\,\frac{1}{s^2+6s+5}$$

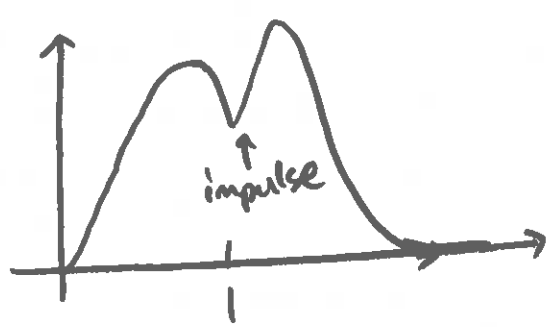
$$= 2\,\frac{1}{(s+5)(s+1)} + e^{-s}\cdot e\,\frac{1}{(s+5)(s+1)}$$

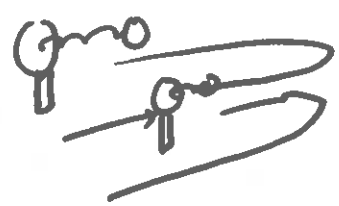
$$\mathcal{L}^{-1}\left\{\frac{1}{s+5}\cdot\frac{1}{s+1}\right\} = e^{-5t} * e^{-t} = \int_0^t e^{-5(t-v)}e^{-v}\,dv$$

$$= \int_0^t e^{-5t+5v-v}\,dv = \int_0^t e^{-5t}e^{4v}\,dv = e^{-5t}\cdot\frac{1}{4}e^{4v}\Big|_{v=0}^{v=t}$$

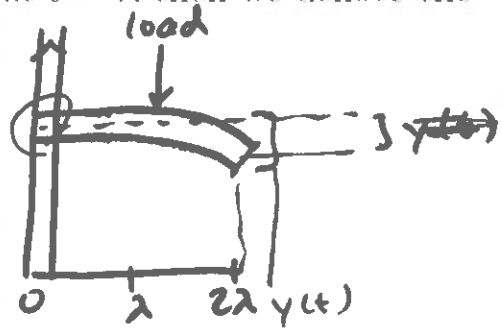
$$= e^{-5t}\left(\frac{1}{4}e^{4t} - \frac{1}{4}\right) = \frac{1}{4}\left(e^{-t} - e^{-5t}\right)$$

so $$y(t) = 2\cdot\left[\frac{1}{4}\left(e^{-t} - e^{-5t}\right)\right] + e\cdot\left[u(t-1)\left[\frac{1}{4}\left(e^{-(t-1)} - e^{-5(t-1)}\right)\right]\right]$$





Example: Suppose a beam of length 2λ is imbedded in a support on one side and free on the other. If a load is acting on the beam at $t = \lambda$ then we denote the amount of deflection away from the norm as $y(t)$.



An equation that models this behavior is

$$EIy^{(4)} = L\delta(t - \lambda)$$

where E is the elasticity of the beam, I is the moment of inertia, and L is the load. Using the initial conditions

to find A & B

$$y(0) = y'(0) = \boxed{y''(2\lambda) = y'''(2\lambda) = 0}$$

solve the IVP.

$$\mathcal{L}\{EIy^{(4)} = L\delta(t-\lambda)\}$$

$$EI\left[s^4Y(s) - s^3\cancel{y(0)}^{0} - s^2\cancel{y'(0)}^{0} - s\underbrace{y''(0)}_{A?} - \underbrace{y'''(0)}_{B?}\right] = Le^{-\lambda s}$$

$$\mathcal{L}\{t^2\} = \frac{2!}{s^3} = \frac{2}{s^3}$$
$$\mathcal{L}\{t^3\} = \frac{3!}{s^4} = \frac{6}{s^4}$$

$$\Rightarrow EI\left[s^4Y(s) - As - B\right] = Le^{-\lambda s}$$

$$\Rightarrow s^4Y(s) - As - B = \frac{L}{EI}e^{-\lambda s}$$

$$\Rightarrow s^4Y(s) = \frac{L}{EI}e^{-\lambda s} + As + B$$

$$\Rightarrow Y(s) = \frac{L}{EI}e^{-\lambda s}\cdot\frac{1}{s^4} + \frac{A}{s^3} + \frac{B}{s^4}$$

$$\Rightarrow Y(s) = \frac{L}{EI}e^{-\lambda s}\cdot\frac{1}{6}\cdot\frac{6}{s^4} + \frac{A}{2}\cdot\frac{2}{s^3} + \frac{B}{6}\cdot\frac{6}{s^4}$$

$u'(t-a) = \delta(t-a)$

$$\Rightarrow y(t) = \frac{1}{6}\cdot\frac{L}{EI}(t-\lambda)^3 u(t-\lambda) + \frac{1}{2}At^2 + \frac{1}{6}Bt^3$$

$$\uparrow$$

$$t^3|_{t\to t-\lambda}$$

$$\Rightarrow y'(t) = \frac{1}{6}\frac{L}{EI}\left[3(t-\lambda)^2 u(t-\lambda) + (t-\lambda)^3\delta(t-\lambda)\right] + At + \frac{1}{2}Bt^2$$

$$\Rightarrow y''(t) = \frac{1}{6}\cdot\frac{L}{EI}\left[6(t-\lambda)u(t-\lambda) + 3(t-\lambda)^2\delta(t-\lambda) \; 3(t-\lambda)^2\delta(t-\lambda) + (t-\lambda)^3\delta'(t-\lambda)\right] + A + Bt$$

$$\Rightarrow y'''(t) = \frac{1}{6}\frac{L}{EI}\left[6u(t-\lambda) + 6(t-\lambda)\delta(t-\lambda) + 6(t-\lambda)\delta(t-\lambda) + 3(t-\lambda)^2\delta'(t-\lambda) + 6(t-\lambda)\delta(t-\lambda) + 3(t-\lambda)^2\delta'(t-\lambda) + 3(t-\lambda)^2\delta'(t-\lambda) + (t-\lambda)^3\delta''(t-\lambda)\right] + 0 + B$$

$y'''(2\lambda) = 0$, $y''(2\lambda) = 0$

$$0 = y'''(2\lambda) = \frac{1}{6}\frac{L}{EI}\left[6\cdot 1 + 0 + \cdots + 0\right] + B$$

$$\Rightarrow \boxed{B = -\frac{L}{EI}}$$

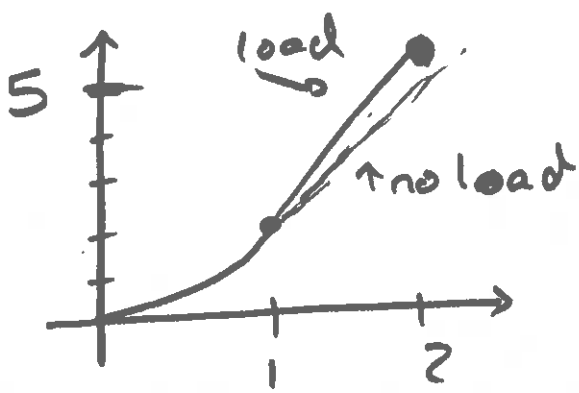
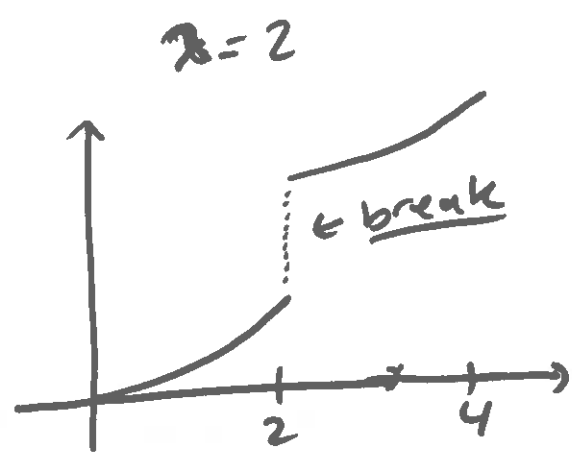
$$0 = y''(2\lambda) = \frac{1}{6}\frac{L}{EI}\left[6\lambda + 0 + \cdots + 0\right] + A + B(2\lambda)$$

$$0 = \frac{1}{6}\frac{L}{EI}[6\lambda] + A - 2\lambda\frac{L}{EI}$$

$$\Rightarrow 0 = A - \lambda\frac{L}{EI} \Rightarrow \boxed{A = \lambda \cdot \frac{L}{EI}}$$

so $$\boxed{y(t) = \frac{1}{6}\cdot\frac{L}{EI}(t-\lambda)^3 u(t-\lambda) + \frac{1}{2}\lambda\cdot\frac{L}{EI}t^2 - \frac{1}{6}\frac{L}{EI}t^3}$$

$$= \begin{cases} \frac{L}{EI}\left[\frac{1}{2}\lambda t^2 - \frac{1}{6}t^3\right], & t < \lambda \\ \frac{L}{EI}\left[\underline{\frac{1}{6}(t-\lambda)^3} + \frac{1}{2}\lambda t^2 - \frac{1}{6}t^3\right], & t > \lambda \end{cases}$$

$L = 6$, $E = I = 1$, $\lambda = 1$ (length 2)

$$\begin{cases} 3t^2 - t^3, & t < 1 \\ 3t - 1, & t > 1 \end{cases}$$